Ligand determinants of physical-validity failure in protein–ligand docking

A re-analysis of the PoseBusters benchmark conditioned on the molecule rather than the method

428 COMPLEXES · 7 DOCKING METHODS · 7,695 POSES

ABSTRACT

Docking methods are judged by whether a predicted ligand pose lands within 2 Å of the crystal structure, but a pose can be accurate and still be physically impossible. The PoseBusters benchmark[1] quantifies this with 18 validity checks and reports failure rates per docking method. It records nothing about the molecules being docked, so it cannot say which ligands are hard.

We computed 27 RDKit descriptors for each of the 428 benchmark ligands, joined them to all 7,695 published poses, and estimated the association between ligand chemistry and each validity failure. Estimates are restricted to ligands capable of failing the check in question, computed per docking method, bootstrapped over ligands rather than poses, and corrected for multiple testing.

Each failure mode has a distinct ligand signature. Strain-energy and internal-clash failures are driven by torsional flexibility (rotatable bonds significant in 4 of 5 methods, odds ratio 1.6–9.5 per standard deviation). Chirality inversion is driven by stereocentre count alone (odds ratio up to 10.4). Cofactor clashes afflict *small*, stereocentre-rich ligands (molecular weight odds ratio 0.17–0.31). The most frequent failure of all — steric clash with the protein — has **no ligand signature**: 5 of 42 coefficients reach significance and none replicates across methods, indicating a property of the method rather than the molecule.

Two boundaries constrain interpretation. Only **41 of 126** method × check strata carry enough failures to estimate at all, and molecular weight and rotatable-bond count are correlated at $\rho \approx 0.73$–0.74 within every method's ligand set, so where both are significant they cannot be separated. Removing complexes whose ligand contacts a crystallographic symmetry mate does **not** reduce protein-clash failure.

We validate the analysis by re-running the checker itself on all 513 crystal structures (99.8% worst-case agreement with the published reference rows) and test generalisation on the 85-complex Astex Diverse Set, where 8 of 14 shortlisted effects replicate with statistical support.

1 Introduction

Molecular docking predicts where a small molecule binds a protein. Performance is conventionally summarised by the fraction of predictions within 2 Å root-mean-square deviation (RMSD) of the crystallographic pose. RMSD is a positional measure: it compares where atoms are, and says nothing about whether their arrangement is chemically possible.

PoseBusters[1] addressed this by pairing RMSD with 18 explicit physical checks — bond lengths and angles within distance-geometry bounds, aromatic rings planar, tetrahedral stereochemistry preserved, no internal or intermolecular steric clashes, internal strain energy within a factor of 100 of a generated conformer ensemble. Applied to seven docking methods, it showed that deep-learning methods with competitive RMSD produce poses that frequently fail these checks, while classical search methods do not.

That result is method-conditioned. The published results table carries a PDB identifier, a chemical-component identifier, a cofactor flag and a sequence identity, and nothing about the molecule's chemistry. The complementary question — which *ligands* break which methods — has not been asked of this data, though the crystal ligands are distributed alongside the results as unanalysed structure files.

We ask it here, and find that the answer is constrained less by the data's size than by collinearity among the descriptors that would answer it.

2 Data and methods

2.1 Data

The PoseBusters paper data[2] comprises 428 protein–ligand complexes released after 2021 (the PoseBusters Benchmark set), 85 complexes from the Astex Diverse Set[3], and a results table giving every validity check and RMSD for seven docking methods — AutoDock Vina, CCDC Gold, DeepDock, DiffDock, EquiBind, TankBind and Uni-Mol — with and without post-prediction energy minimisation. The predicted poses themselves were not released, so the published check results are the analysis input. Unless stated, results are for raw (un-minimised) poses on the 428-complex benchmark set.

Two encoding properties of that table govern everything downstream. A blank check value means the check *was not run*, not that it failed; both flatness columns are blank for all 2,996 minimised rows. And 85 rows have every check blank because the method produced no pose at all — treating those as passing inflates validity rates. Both are handled explicitly.

2.2 Descriptors

27 descriptors were computed per crystal ligand with RDKit[4]: size (molecular weight, heavy atoms), flexibility (rotatable bonds, fraction sp^3, amide bonds), ring system (total, aromatic, aliphatic, largest ring,

spiro and bridgehead atoms), stereochemistry (tetrahedral centres, stereogenic double bonds), and polarity and composition (TPSA, cLogP, donors, acceptors, formal charge, heteroatoms, halogens). All 513 ligands parsed and sanitised without error. Descriptors describe the crystal ligand, so they are properties of the molecule that was docked, independent of which method produced a pose.

2.3 Eligibility

A ligand with no stereocentres cannot fail the tetrahedral-chirality check; one with no aromatic ring cannot fail aromatic-ring flatness. Including such ligands in a check's denominator deflates its failure rate and inflates any effect measured against the gating descriptor, because the comparison then encodes whether the check *can* fire rather than whether the molecule induces failure. Each check is therefore gated on the count of the substructure it inspects, using the checker's own SMARTS patterns where applicable. Gating raises the observed chirality-failure rate from 21.4% to 36.7% and reduces its association with stereocentre count from $d = 1.09$ to $d = 0.35$.

The four cofactor-clash checks are conditioned differently: they pass almost automatically on complexes that contain no cofactor (2.4% failure versus 22.5% where one is present), so they are restricted to cofactor-bearing complexes, which is a property of the receptor rather than the ligand.

2.4 Estimation

Effect sizes are Cohen's d between failing and passing poses, estimated separately for each docking method — the methods differ by more than an order of magnitude in competence, so pooling them describes the weakest as much as the strongest. Confidence intervals come from a bootstrap that resamples *ligands*, not poses: each ligand contributes one pose per method, so treating 2,140 rows as independent observations of 428 ligands would report intervals roughly $\sqrt{5}$ too narrow. Strata with fewer than 15 failures or fewer than 30 distinct ligands are not estimated. Across the full grid of estimates, p-values are corrected by the Benjamini–Hochberg procedure[5] at $\alpha = 0.05$, treating every estimate as one family.

Because marginal comparisons cannot separate correlated descriptors, each check is also modelled by logistic regression on a decorrelated descriptor basis with standard errors clustered on ligand. Where the marginal estimates and the models disagree, the models govern.

2.5 Crystal contacts

The benchmark's journal version reports on 308 of the 428 complexes, excluding those whose ligand contacts a crystallographic symmetry mate on the grounds that such contacts could inflate protein-clash failures. That subset was not published. We recomputed it from first principles with gemmi[6]: each deposited structure was expanded by its spacegroup and the closest approach measured between the ligand of interest and any symmetry image of the protein, counting only amino-acid residues. 104 of 427 resolvable complexes fall within 4.0 Å; a broader criterion including solvent and cryoprotectant flags 119. These bracket the 120 the authors removed, and are an independent estimate rather than a reproduction of their list.

3 Results

3.1 Accuracy and validity diverge sharply by method class

Classical search methods produce accurate poses about half the time, and nearly all of those poses are physically valid. Deep-learning methods produce fewer accurate poses, and most of the accurate ones are invalid (Table 1). DiffDock is the strongest deep-learning method by RMSD alone and 64.2% of its accurate poses fail at least one validity check.

TABLE 1 Accuracy, validity, and their intersection, per method. Raw poses, 428 complexes. The final column is the share of RMSD-accurate poses that are physically invalid.

METHOD	CLASS	RMSD ≤ 2 Å	VALID	BOTH	OF ACCURATE POSES, INVALID
vina	classical	52.3%	97.4%	51.2%	2.2%
gold	classical	51.2%	93.9%	47.7%	6.8%
diffdock	deep learning	37.9%	24.1%	13.6%	64.2%
unimol	deep learning	22.9%	5.6%	1.9%	91.8%
deepdock	deep learning	17.8%	8.2%	4.7%	73.7%
tankbind	deep learning	15.0%	4.7%	2.6%	82.8%
equibind	deep learning	2.6%	0.9%	0.2%	90.9%

3.2 The published checks reproduce

Every result here reads the published check outcomes rather than recomputing them, so the analysis depends on interpreting those columns correctly. We tested that by re-running PoseBusters on all 513 crystal ligands against their own receptors and comparing to the paper's own reference rows. Worst-case agreement across 17 mapped checks is `99.81%`, with 15 of 17 checks agreeing on all 513 structures. The two disagreements are single structures on different checks — `7V3S` on aromatic ring flatness and `7T0U` on strain energy, the latter one of the two borderline failures the paper itself reports.

3.3 Two thirds of the grid is unmeasurable

Of 126 method × check combinations, 41 carry enough failures and enough distinct ligands to estimate an effect (Table 2). The remaining 85 are not null results; they are unmeasured, mostly because the check never failed for that method. Across the estimable strata, 287 descriptor associations were computed and **95 survive** false-discovery-rate correction.

TABLE 2 Estimability of the 126 method × check strata. Thresholds: at least 15 failures and at least 30 distinct ligands.

OUTCOME	STRATA	SHARE
Check never failed for that method	59	47%
Estimated	41	33%
Fewer than 15 failures	25	20%
Fewer than 30 distinct ligands	1	1%

3.4 Surviving associations are dominated by size and flexibility

The largest surviving effects (Figure 1, Table 3) concentrate on internal-geometry checks — internal clashes, bond lengths, bond angles, strain energy — and on two descriptors, molecular weight and rotatable-bond count. Both appear repeatedly on the same check for the same method with overlapping intervals, which is not two findings but one signal counted twice: the two descriptors correlate at ρ ≈ 0.73–0.74 within every method's ligand set. Marginal estimates cannot separate them.

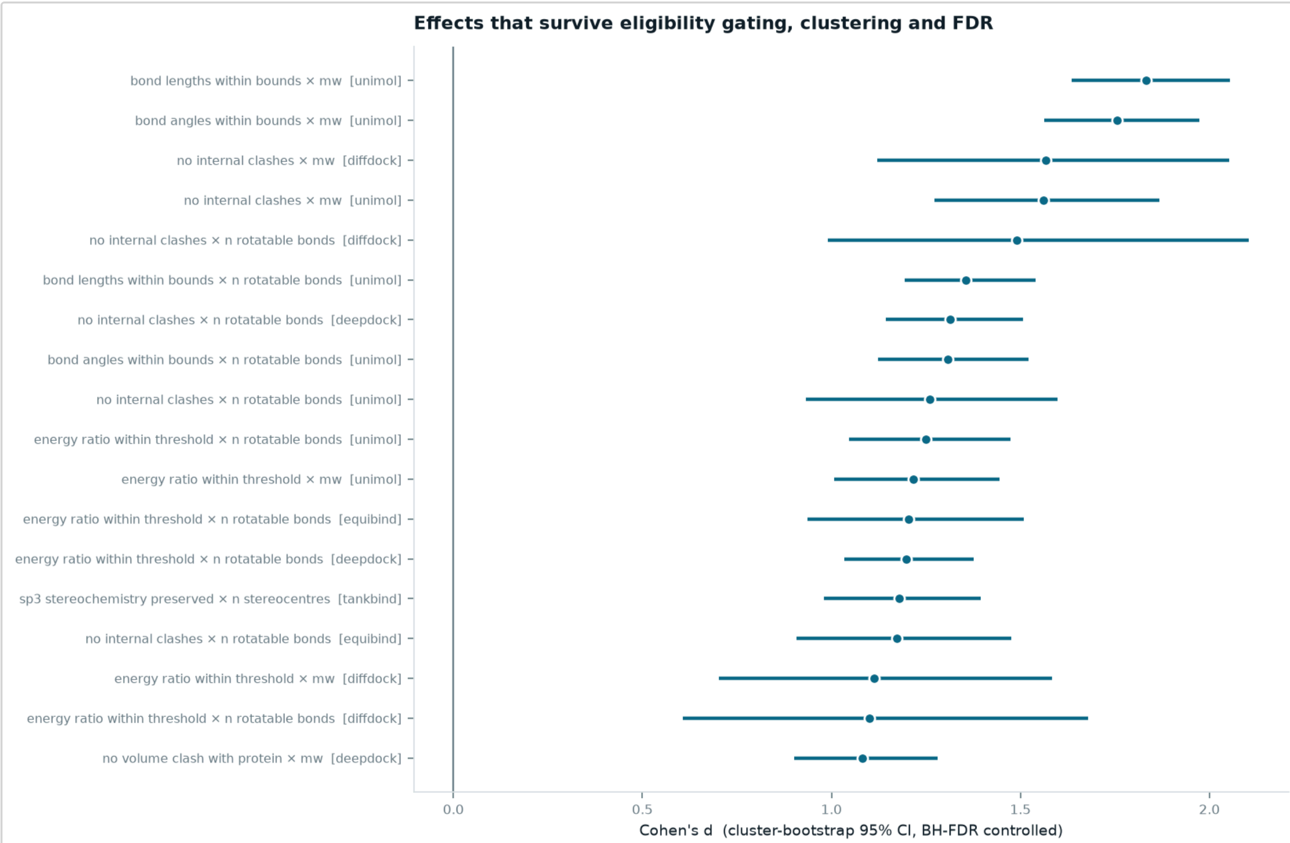

FIGURE 1 The 18 largest surviving marginal associations with cluster-bootstrap 95% confidence intervals, after eligibility gating and Benjamini–Hochberg correction. Molecular weight and rotatable-bond count recur on the same check for the same method, reflecting their collinearity rather than independent effects.

TABLE 3 Twelve largest surviving associations by |d|. Failures and ligands give the denominator each estimate rests on.

METHOD	CHECK	DESCRIPTOR	D	95% CI	FAILURES	LIGANDS
unimol	bond lengths within bounds	mw	1.83	[1.64, 2.05]	269	427
unimol	bond angles within bounds	mw	1.76	[1.56, 1.97]	284	427
diffdock	no internal clashes	mw	1.57	[1.12, 2.05]	25	426
unimol	no internal clashes	mw	1.56	[1.27, 1.87]	66	427
diffdock	no internal clashes	n rotatable bonds	1.49	[0.99, 2.10]	25	426
unimol	bond lengths within bounds	n rotatable bonds	1.36	[1.19, 1.54]	269	427
deepdock	no internal clashes	n rotatable bonds	1.31	[1.14, 1.51]	205	427
unimol	bond angles within bounds	n rotatable bonds	1.31	[1.12, 1.52]	284	427
unimol	no internal clashes	n rotatable bonds	1.26	[0.93, 1.60]	66	427
unimol	energy ratio within threshold	n rotatable bonds	1.25	[1.05, 1.47]	165	427
unimol	energy ratio within threshold	mw	1.22	[1.01, 1.45]	165	427
equibind	energy ratio within threshold	n rotatable bonds	1.20	[0.94, 1.51]	92	426

3.5 Each failure mode has a distinct ligand signature

Table 4 is the analysis's central result: for each check, the ligand properties that carry independent signal once the other six descriptors are held fixed, and how many docking methods agree. Four patterns emerge.

Strain energy and internal clashes are flexibility phenomena. Rotatable-bond count is significant for both in 4 of 5 methods with consistent sign (odds ratios 1.6–9.5 per standard deviation). Molecular weight is also significant for internal clashes in 3 of 5, but DeepDock's coefficient is inverted, so size is not consistent where flexibility is. **Chirality inversion is a stereocentre-count phenomenon** and nothing else, with the largest effect in the study — an odds ratio of 10.4 for TankBind. **Cofactor clashes afflict small, stereocentre-rich ligands**: within the 187 cofactor-bearing complexes, molecular weight has an odds ratio of 0.17–0.31 in 4 of 5 methods, meaning smaller ligands clash more, while stereocentre count raises failure by 2.3–3.2×.

The most frequent failure has no ligand signature at all. Steric clash with the protein fails on roughly 84% of deep-learning poses, yet only 5 of 42 coefficients across 6 methods reach significance and no descriptor agrees on more than two. The dominant failure mode is not a property of the molecule. The failures that *do* discriminate between ligands — strain, self-clash, chirality — are predictable from three descriptors.

3.6 Flexibility versus size is not separable where both appear

The natural hypothesis is that torsional freedom, not molecular size, drives strain and clash failures: a flexible molecule has more ways to be locally wrong while remaining globally close to the crystal pose. Testing it requires holding one descriptor fixed while varying the other, which their correlation forbids in a stratified table. Logistic models with both descriptors entered together give a per-method answer that does not agree with itself (Table 5).

For **DiffDock**, neither descriptor reaches significance once the other is held fixed ($p = 0.941$ and 0.108), despite both surviving correction as marginal effects. EquiBind gives the only clean flexibility-only result and is the weakest method in the benchmark. DeepDock, TankBind and Uni-Mol show both descriptors carrying independent signal; DeepDock's weight coefficient is negative, the opposite sign to a naive "larger molecules fail more" account. Gold and Vina fail too rarely to fit.

The contrast is therefore not identified in this dataset for the method it was posed about. This is a property of the benchmark's ligand set, not of the sample size: the correlation is essentially constant across every method's stratum, so no reweighting or larger subset of these complexes would separate the two.

TABLE 4 Ligand properties carrying independent signal per check, from the clustered logistic models. Odds ratio per standard deviation, other descriptors held fixed; below 1 means the failure becomes less likely as the property rises. Only descriptors significant for at least two methods are listed.

CHECK	LIGAND PROPERTY	DIRECTION	METHODS	ODDS RATIO / SD
energy ratio within threshold	n rotatable bonds	↑ raises failure	4 of 5	2.08–7.90
no clashes with organic cofactors	mw	↓ lowers failure	4 of 5	0.17–0.31
no clashes with organic cofactors	n stereocentres	↑ raises failure	4 of 5	2.28–3.15
no internal clashes	n rotatable bonds	↑ raises failure	4 of 5	1.55–9.53
no volume clash with organic cofactors	mw	↓ lowers failure	4 of 5	0.15–0.35
no clashes with organic cofactors	clogp	↑ raises failure	3 of 5	1.54–2.32
no internal clashes	mw	↑ raises failure	3 of 5	2.70–3.35
bond lengths within bounds	mw	↑ raises failure	2 of 2	3.33–47.06
energy ratio within threshold	mw	↑ raises failure	2 of 5	2.14–3.55
no clashes with organic cofactors	n rotatable bonds	↑ raises failure	2 of 5	1.96–2.29
no clashes with protein	clogp	↓ lowers failure	2 of 6	0.56–0.67
no internal clashes	formal charge	↓ lowers failure	2 of 5	0.73–0.82
no internal clashes	n stereocentres	↑ raises failure	2 of 5	1.66–1.79
no volume clash with organic cofactors	n stereocentres	↑ raises failure	2 of 5	3.11–3.55
no volume clash with protein	clogp	↓ lowers failure	2 of 5	0.57–0.65
sp3 stereochemistry preserved	n stereocentres	↑ raises failure	2 of 3	4.33–10.43

TABLE 5 Clustered logistic models of the strain-energy check, both descriptors entered together with five others. Standard errors clustered on ligand.

METHOD	P (MW)	P (ROTATABLE BONDS)	VERDICT
diffdock	0.941	0.108	Neither — not resolvable
equibind	0.956 (negative)	0.000	Flexibility only
deepdock	0.020 (negative)	0.000	Both carry independent signal
tankbind	0.044	0.001	Both carry independent signal
unimol	0.010	0.002	Both carry independent signal
gold	0.471 (negative)	0.395	Neither — not resolvable
vina	–	–	No model fit (too few failures)

3.7 Crystal contacts do not inflate protein-clash failure

Protein clash is the most-failed check for deep-learning methods, so if it concentrated in complexes whose ligand touches a symmetry mate, the headline failure rate would be partly an artifact of crystal packing. It does not (Table 6). One method of seven shows a significant difference and its clash failure is *lower* among contact-bearing complexes, the reverse of the predicted direction. The remaining six are indistinguishable from zero, and the two classical methods fail so rarely — Vina on 2 of 323 no-contact poses, Gold on 16 of 322 — that their point estimates carry no information.

TABLE 6 Protein-clash failure by crystal-contact status, with cluster-bootstrap intervals on the difference (2,000 replicates, resampling complexes).

METHOD	CONTACT	NO CONTACT	DIFFERENCE (PP)	95% CI	EXCLUDES ZERO
unimol	68.9% (103)	86.1% (323)	−17.1	[−27.0, −7.7]	yes
diffdock	61.2% (103)	70.5% (322)	−9.3	[−19.6, +1.8]	no
tankbind	77.9% (104)	85.1% (323)	−7.3	[−16.2, +1.6]	no
equibind	96.1% (103)	98.4% (322)	−2.3	[−6.6, +1.2]	no
gold	2.9% (102)	5.0% (322)	−2.0	[−5.9, +2.4]	no
vina	0.0% (104)	0.6% (323)	−0.6	[−1.5, +0.0]	no
deepdock	91.3% (103)	85.1% (323)	+6.1	[−0.9, +12.5]	no

3.8 Invalidity is not confined to marginal poses

Reporting invalidity only among RMSD-accurate poses conditions on an outcome that shares an upstream cause with validity, which can induce association between otherwise unrelated quantities. Banding raw RMSD avoids this and uses every pose (Table 7). Deep-learning validity is already poor at sub-Ångström accuracy — DiffDock 61.3%, Uni-Mol 33.3% — and does not recover at larger RMSD. The classical methods are above 88% in every band.

Conditioned on accuracy, the invalid share rises monotonically with rotatable-bond count in both method classes, and is close to flat against ring count (Figure 2). Per-check, the strain-energy and internal-clash failures climb steeply with flexibility while planarity checks barely move (Figure 3) — consistent with the surviving associations in §3.4, and subject to the same identifiability limit in §3.5.

TABLE 7 Share of poses passing all applicable validity checks, by RMSD band. Unconditioned on accuracy.

METHOD	(0.0, 1.0]	(1.0, 2.0]	(2.0, 3.0]	(3.0, 5.0]	(5.0, 10.0]	(10.0, INF]
deepdock	55.0%	16.1%	12.5%	6.8%	1.1%	0.0%
diffdock	61.3%	20.0%	20.0%	17.8%	18.6%	12.7%
gold	95.2%	88.9%	95.0%	90.7%	97.6%	100.0%
tankbind	36.4%	13.2%	3.0%	1.8%	0.0%	6.9%
unimol	33.3%	2.5%	6.2%	5.2%	4.2%	0.0%
vina	100.0%	93.5%	97.6%	96.2%	96.2%	100.0%

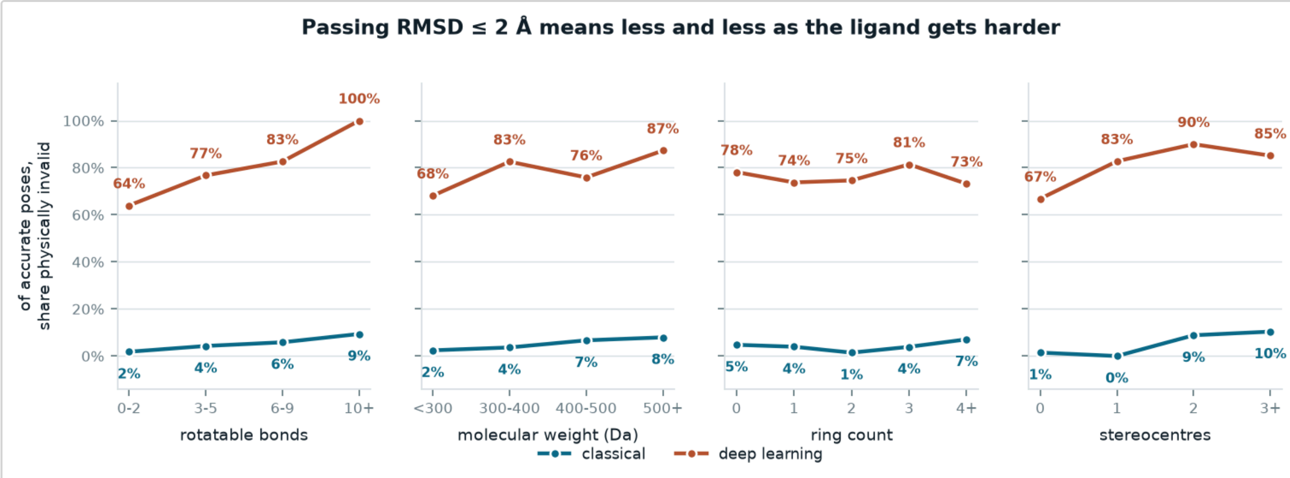

FIGURE 2 Share of RMSD-accurate poses that are physically invalid, across four ligand partitions. Monotone against rotatable-bond count in both method classes; close to flat against ring count. Correlated descriptors mean this does not by itself establish which property is responsible.

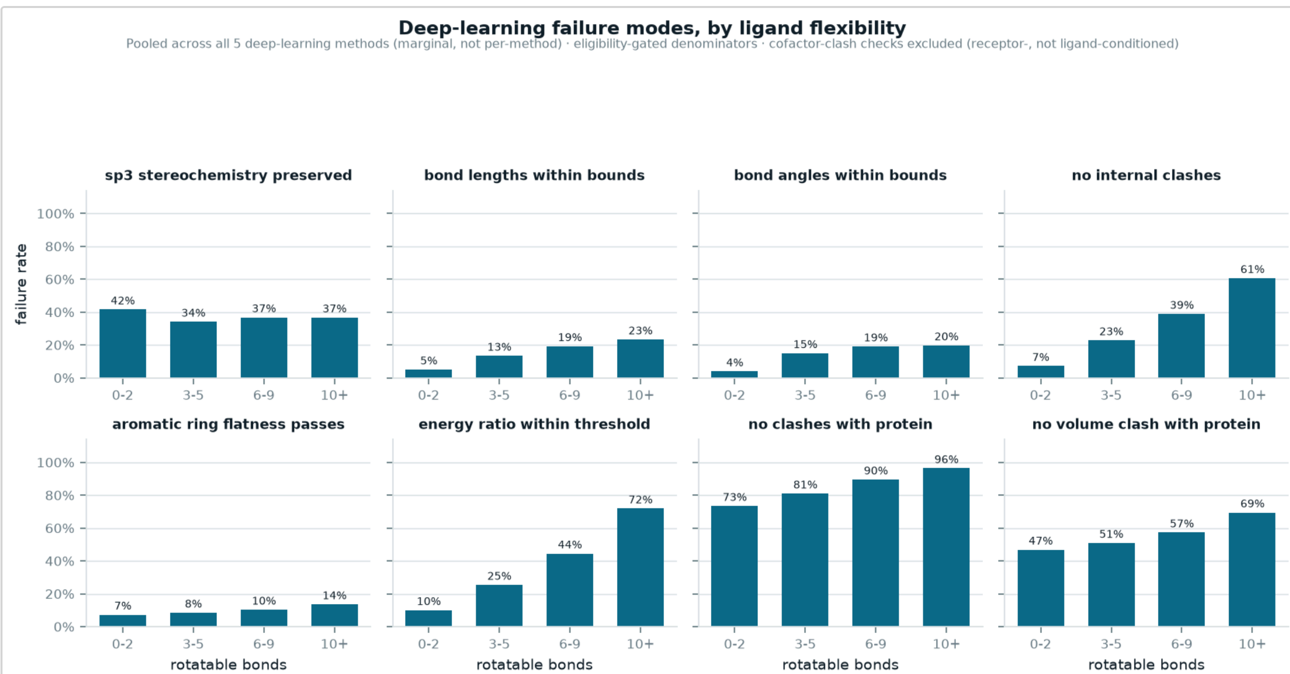

FIGURE 3 Per-check failure rate against rotatable-bond count, pooled across the five deep-learning methods, with eligibility-gated denominators. Cofactor-clash checks are excluded as receptor- rather than ligand-conditioned.

3.9 Replication on a held-out set

All estimates above were both discovered and computed on the same 428 complexes. The 85 Astex complexes were untouched throughout, so re-estimating a shortlist there is an independent test. Of 14 checkable triples, **8 replicate with statistical support** — both intervals excluding zero (Table 8). 13 of 14 agree in sign, but agreement between two estimates that both straddle zero carries no information, so 8/14 is the defensible figure. The strain-energy and internal-clash effects against rotatable bonds replicate for every deep-learning method tested; the stereochemistry and protein-clash effects largely do not.

TABLE 8 Shortlisted effects re-estimated on the held-out Astex Diverse Set. Rows resting on fewer than 15 failures on either side are flagged thin.

METHOD	CHECK	DESCRIPTOR	D (BENCHMARK)	D (ASTEX)	REPLICATED	THIN
diffdock	no clashes with protein	n rotatable bonds	0.60	0.30	no	
unimol	energy ratio within threshold	n rotatable bonds	1.25	0.96	**yes**	
unimol	no internal clashes	n rotatable bonds	1.26	0.87	**yes**	thin
unimol	sp3 stereochemistry preserved	n stereocentres	0.84	0.25	no	
unimol	no clashes with protein	n rotatable bonds	0.54	0.61	**yes**	
deepdock	energy ratio within threshold	n rotatable bonds	1.20	1.38	**yes**	
deepdock	no internal clashes	n rotatable bonds	1.31	1.14	**yes**	
deepdock	sp3 stereochemistry preserved	n stereocentres	0.05	0.03	no	
deepdock	no clashes with protein	n rotatable bonds	0.92	0.65	**yes**	
tankbind	energy ratio within threshold	n rotatable bonds	0.97	1.03	**yes**	
tankbind	no internal clashes	n rotatable bonds	0.86	0.47	**yes**	
tankbind	sp3 stereochemistry preserved	n stereocentres	1.18	0.17	no	
tankbind	no clashes with protein	n rotatable bonds	0.37	0.22	no	
gold	no clashes with protein	n rotatable bonds	0.15	−1.00	no	thin

4 Discussion

The clearest result is negative, and it is about the benchmark rather than the methods. The two descriptors that any account of docking failure would want to distinguish — how big a molecule is and how many ways it can bend — are entangled throughout this ligand set at a level that no analysis of these 428 complexes can undo. Marginal estimates make both look significant; multivariate models make neither look significant for the method the question is most often asked about. Answering it would require a benchmark assembled to decorrelate them, containing large rigid ligands and small flexible ones in comparable numbers.

The crystal-contact result is similarly cautionary. The exclusion of contact-bearing complexes is a reasonable methodological precaution, but in this data it does not do what it is assumed to do: protein-clash failure is no higher among the complexes it removes. That is worth knowing before treating the 308-complex subset as a cleaner benchmark than the 428.

What does survive is coarser than the descriptor-level story the marginal grid appears to tell. Deep-learning poses fail internal-geometry checks — clashes, bond geometry, strain — at rates that rise with molecular size and flexibility jointly, and the failure is not confined to poses that barely clear the RMSD threshold. It is present at sub-Ångström accuracy, which suggests these methods are not producing slightly-imperfect physical molecules but rather point clouds that happen to be near the right coordinates.

Finally, 85 of 126 strata could not be estimated at all, most because the check never failed for that method. Reporting an effect grid without that denominator invites reading absence of estimate as absence of effect.

5 Limitations

Descriptors describe the docked molecule, not the returned pose, so nothing here measures how wrong a pose is beyond the binary checks. The predicted poses were never released, so the methods' check results cannot be independently recomputed; only the crystal-structure rows could be, and were (§3.2). The crystal-contact flag is our own estimate and brackets rather than reproduces the authors' undisclosed subset. Several surviving associations rest on small failure counts even after gating — DiffDock's internal-clash effects on 25 failures — and single rows should not be read as decisive. The cofactor-clash checks are conditioned on the receptor and are reported only in the estimable-strata counts, not interpreted as ligand chemistry.

6 Data and code availability

Source data are the PoseBusters paper data[2] on Zenodo and deposited structures from the RCSB PDB[7], both fetched automatically by the pipeline. Every table in this paper is read from the analysis outputs at render time and every figure is inlined from the generated originals, so the document cannot diverge from the analysis it reports. The pipeline is deterministic under fixed seeds and covered by 37 tests.

References

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Analysis conditioned on ligand chemistry rather than docking method. All estimates carry cluster-bootstrap intervals; no point estimate is reported without one. Generated from `reports/tables/` — regenerate with `python -m pb.paper_html`.